Mentor Session Plan - MongoDB and Mongoose with AI-Assisted Development

Total Duration: 3 hours

Learning Objectives

- Understand MongoDB schema design and relationships in Mongoose
- Write Mongoose queries for common database operations (CRUD, filtering, aggregation)
- Understand the difference between System Prompts and User Prompts
- Manage the Context Window effectively during AI interactions
- Use Agent Mode for implementing database models, generating test data, seed scripts, and test scripts
- Use Plan Mode for brainstorming and designing database schemas

Session Schedule

Time	Topic	Detail
~20 min	Intro + Project Setup	Introduce the ThreadHive app and its structure (models, data, populate_db.js, queries.js). Verify MongoDB Atlas setup, install dependencies, and seed the database. Review the three schemas (User, Subreddit, Thread) and their relationships.
~60 min	Manual Query Implementation	Live-code the 12 manual implementation queries in `queries.js`. Demo different query types: find by field, filtering, date ranges, CRUD operations, aggregation. Learners follow along and implement queries with guidance.
~20 min	AI Concepts: Prompts and Context Window	Explain System Prompts vs User Prompts with examples. Demo Context Window tracking in Copilot Chat. Cover practical context management tips (new chat per task, attach only relevant files, Plan before Agent).
~60 min	AI Modes: Plan and Agent	Demo AI use cases. **Agent Mode** : (A) Recreate Mongoose models, (B) Generate seed script, (C) Generate test script. **Plan Mode** : (D) Brainstorm features with MoSCoW prioritization, (E) Design MongoDB schema.
~15 min	Best Practices and Wrap-up	Review "trust but verify" principle for AI outputs. Emphasize model selection for Plan Mode (use strong reasoning models). Reiterate context management habits. Q&A and preview next week's topics.

Mentor Tips

Intro + Project Setup (~20 min)

- Do a quick check to confirm prerequisites:
  - MongoDB Atlas account with a free cluster set up
  - MongoDB connection string setup in `.env` file
- Common MongoDB Atlas connection issues to address in tips (not main session time):
  - **Whitelist IP address**: Ensure "0.0.0.0/0" is whitelisted in Network Access (for development only)
  - **Database user credentials**: Verify the username and password are correct and user has read/write permissions
  - **Connection string format**: Make sure to replace `<password>` placeholder and specify database name
  - **Network restrictions**: Check if learner's network/firewall blocks MongoDB ports
- After running `npm run populate`, have learners verify the data in MongoDB Atlas (Browse Collections) to confirm seeding worked.
- Spend 5 minutes carefully reviewing the schemas in `models/` - emphasize the relationships (Thread references User and Subreddit; Thread has numeric upvotes/downvotes fields and a computed voteCount).

Manual Query Implementation (without Copilot) (~60 min)

- Live-code each query type yourself first, explaining the Mongoose syntax as you go.
- Most learners will be new to Mongoose syntax - take time to explain the queries.
- Keep track of time. If short on time, you can use Copilot's inline code suggestions to speed up some of the queries.

AI Concepts: Prompts and Context Window (~20 min)

- Keep the System Prompt vs User Prompt explanation concise but clear.
- For the Context Window exercise, have learners open the model selector in Copilot Chat and go to "Manage Models":
  - Point out the context window sizes (e.g., Claude Sonnet 4.5: 200K tokens, GPT-4o: 128K tokens)
  - Show the context usage indicator at the top right of chat during a conversation
- Emphasize the three key context management tips:
  1. **Start a new chat for each new task** - don't carry over unrelated context
  2. **Only attach files directly relevant to the current task** - resist the urge to attach everything
  3. **Use Plan Mode before Agent Mode for large tasks** - planning in a focused chat keeps context clean before execution

AI Modes: Plan and Agent (~60 min)

- Start with the three **Agent Mode** use cases where learners actively participate:
  - **Use Case A — Recreate Mongoose Models:** Emphasize that learners must commit their Part 1 work before deleting the `models/` and `data/` files. After AI generates the models, have learners verify each field against the schema tables in the MLS. This is a natural moment to reinforce the "trust but verify" principle.

- **Use Case B — Generate Seed Script:** Walk through the prompt structure and review the generated data for referential integrity. Have learners run the seed script and verify in Atlas.
- **Use Case C — Generate Test Script:** Highlight that this is a standalone Node.js script, not a test framework. Review the generated tests for completeness. We are not covering unit / integration testing frameworks here — that is covered in the weeks to come.

- Then transition to **Plan Mode** for the two demo-only use cases (learners observe, not practice):
    - Introduce Plan Mode as a "thinking space" before implementation.
    - Use Plan Mode when: brainstorming, designing schemas, planning refactoring, preparing test strategies.
    - Always use a **strong reasoning model** in Plan Mode (Claude Opus 4.5+, GPT-5.2+, Gemini 2+).
    - Review and refine the plan before switching to Agent Mode.
    - **Use Case D — Brainstorm Features:** Demo the MoSCoW prioritization prompt and discuss the quality of the output.
    - **Use Case E — Design MongoDB Schema:** Demo the schema design prompt with `features.md` attached. Discuss the two schema approaches and trade-offs.
    - Emphasize that use case D and E are demo-only to illustrate the power of Plan Mode for brainstorming and design. We will not be implementing these features or schemas in this session.

Best Practices and Wrap-up (~15 min)

- Reinforce the "trust but verify" principle:
    - Always review generated code for logic errors, security issues, and best practices
    - Run and test the code - don't just copy-paste
- Emphasize model selection strategy:
    - Use **strong reasoning models** for Plan Mode and **faster models** (Sonnet 4.5+, GPT-4o) for Agent Mode implementation tasks
- Remind learners to monitor their premium request usage to avoid running out mid-month
- Reiterate the key context management habits:
    - New chat per task
    - Attach only relevant files
    - Plan before Agent for complex work